



FXD Engineering Project Binder

Volume 1 - Milestones 01-25

Audited engineering record based on the approved documentation-recovery structure

Project	FXD - Intelligent Industrial Fixture Design
Repository	kool1160/fxd-fixture-design
Document Class	Engineering development history / professional project binder
Volume Status	Closed audited record through Milestone 25
Audit Basis	Merged pull requests, tests, CI, project law, architecture, decision, and project records
Runtime Changes	None - documentation only

AI proposes. Engineering validates.

The software should think like an experienced fixture engineer.

1. Executive Summary

Volume 1 records FXD's progression from a dependency-free geometry proof into a CAD-neutral digital fixture engineer with real STEP/OCP geometry, deterministic locating and weld-fixture rules, explainable concept generation, validation gates, visual engineering review, controlled edits, complete fixture structures, manufacturing geometry, drawings, and cost/volume/manufacturability analysis.

Governing principles

- Source customer CAD remains immutable.
- Deterministic geometry, mathematics, and engineering rules remain authoritative.
- AI is advisory and may not approve a fixture.
- Every recommendation and validation result must remain explainable and traceable.
- Manufacturing practicality and qualified human approval remain mandatory.

2. Master Milestone Recovery Tracker

Exact historical test counts are reported only when the controlling repository evidence states them.

M	Milestone	Status / PR	Scope Summary	Testing / Evidence	Boundary
1	Establish the Runnable Technical Baseline	Complete #2	Created the dependency-free geometry proof, explicit millimetre units, transforms, intersection and clearance checks, deterministic neutral serialization, CI entry points, and the first CAD-neutral architecture boundary.	Baseline geometry and unit tests passed; the proof deliberately used bounded review geometry rather than claiming production B-Rep behavior.	No STEP parser, real topology, production fixture geometry, certification, or production release.
2	Import STEP Assemblies into a Normalized Product Model	Complete #3	Added immutable CAD-neutral assembly hierarchy, repeated instances, nested transforms, units, bounds, topology summaries, source hashing, and explicit import failures.	Hierarchy, transform, unit, repeated-instance, source-hash, and failure-path coverage passed.	The normalized model did not modify source CAD or claim a full ISO 10303 implementation.
3	Build the Engineering-Annotation Workflow	Complete #4	Created source-bound annotations for build orientation, loading direction, process, quantity, critical characteristics, locating permissions, forbidden contacts, weld joints, shop constraints, and visible assumptions.	Deterministic round-trip, source binding, reference validation, and explicit-unknown behavior passed.	Annotations describe manufacturing intent; they do not prove or approve a fixture.
4	Generate Baseplate, Supports, Stops, and Locator Primitives	Complete #5	Generated traceable parameterized baseplates, supports, stops, pin envelopes, and locator primitives with initial overlap, forbidden-contact, missing-intent, and trapped-part findings.	Primitive identity, parameter, overlap, forbidden-contact, missing-intent, and trapped-part coverage passed.	Proof primitives only; no complete structure, structural adequacy, or production-ready geometry.
5	Create and Rank Complete Fixture Concepts	Complete #6	Produced deterministic alternatives for minimum cost, fast loading, and high repeatability with locating/clamping rationale, warnings, score components, and copy-on-write corrections.	Concept generation, ranking, score decomposition, rationale, alternatives, and correction behavior passed.	Relative engineering ranking only; not supplier quotation, process certification, or production approval.
6	Model Weld, Operator, and Robot Access	Complete #7	Added weld-approach, manual-tool, robot, operator, loading, and unloading envelopes with deterministic obstruction findings and visible uncertainty.	Envelope generation, obstruction findings, incomplete-evidence handling, and traceability passed.	No robot path planning, ergonomic certification, or weld-process approval.
7	Add Standard Clamp and Tooling Libraries	Complete #8	Implemented vendor-neutral clamp, pin, and rest contracts and deterministic selection using force, stroke, mounting, envelope, licensing, and review limitations.	Library identity, suitability, force/stroke, mounting, envelope, fallback, and licensing metadata passed.	Review data only; not certified supplier selection or commercial-CAD equivalence.
8	Export a Fabrication-Ready Review Package	Complete #9	Added deterministic review-package export with STEP-shaped proof geometry, DXF profiles, BOM, setup instructions, manifest, validation findings, units, revision, and engineering-review labels.	Package determinism, manifest linkage, units, revision identity, and review-status coverage passed.	Review package only; proof outputs were not released production drawings.
9	Capture Engineer Corrections and Reusable Knowledge	Complete #10	Added attributable copy-on-write corrections, local private storage, sanitized training export, decision gating, and protections against silently promoting preferences into universal rules.	Correction attribution, revision, privacy, sanitized export, and promotion-gating coverage passed.	No autonomous engineering-rule rewriting.
10	Add CAD-Specific Connectors	Complete #11	Established optional thin CAD connectors around the neutral model, including neutral STEP translation, review-package export, conservative SOLIDWORKS probing, and approval-gated destructive operations.	Connector routing, capability probing, neutral handoff, failure behavior, and approval gates passed.	The engineering core remained independent of SOLIDWORKS and every single CAD vendor.

M	Milestone	Status / PR	Scope Summary	Testing / Evidence	Boundary
11	Integrate a Real Geometry Kernel	Complete #12	Integrated pinned cadquery-ocp behind the CAD-neutral kernel boundary with real STEP import/export, XCAF hierarchy, topology, stable references, Booleans, distance, clearance, malformed-input handling, source immutability, and licensing records.	Real-kernel import, topology, references, Booleans, distance, clearance, malformed input, source immutability, and licensing passed; superseded PR #13 was not controlling evidence.	Kernel capability does not certify fixture geometry or manufacturing readiness.
12	Complete the Deterministic Locating and Constraint Solver	Complete #14	Implemented six-degree-of-freedom locating analysis using contact points, normals, roles, rigid-body constraint rows, rank analysis, and physical DOF classification.	Rank, DOF classification, invalid-reference, redundancy, underconstraint, and recommendation-blocking coverage passed.	Rigid-body locating analysis only; no elastic deformation, complete tolerance stack, or structural simulation.
13	Complete Weld-Fixture Engineering Rules	Complete #15	Added deterministic weld-process reasoning for process, direction, sequence, heat input, distortion, tack, release, support placement, and clamp-force direction with traceable rule identities.	Clamp direction, tack/release, sequence, evidence, and rule-traceability coverage passed.	No thermal FEA, weld-quality certification, or WPS approval.
14	Establish the Manufacturing-Aware Geometry Foundation	Complete #16/#17/#19/#22	Added manufacturing method, material, thickness, fit, clearance, allowance, interface, and operation metadata; deterministic cut-operation planning; source binding; feature-order validation; and STEP/DXF parity.	STEP/DXF parity, wrong source hash, reordered features, malformed outputs, substitution, and pinned-OCF acceptance infrastructure passed.	Foundation closure did not itself claim application-grade visual acceptance; that was completed under Milestone 17.
15	Build the Full Deterministic Validation Pipeline	Complete #21	Unified geometry, locating, access, tooling, tolerance, and manufacturing evidence into a versioned ValidationResult that gates export and fails closed for missing, invalid, mismatched, unitless, unsupported, or empty evidence.	Identity matching, evidence digest, intended interfaces, unrelated collisions, locating/clamp adequacy, provisional findings, and export blocking passed.	Validation supports qualified review; it is not certification.
16	Build the First Serious Visual Engineering Application	Complete #24	Added a local engineering application for STEP import, rotatable review, layers, assumptions, findings, corrections, approval decisions, and neutral project save/reload with stale and invalid approval blocking.	Annotation preservation, immutable source identity, evidence persistence, tamper detection, unsafe approval blocking, edit revocation, and unknown-feature rejection passed.	Initial display used normalized review geometry; production-quality B-Rep display followed in Milestone 17.
17	Prove and Expose Real-Kernel Geometry	Complete #28	Exposed real product and fixture B-Rep geometry with tessellation, selectable faces/edges, sections, feature findings, collision highlighting, wireframe, transparency, layers, regeneration, and reconstruction from embedded STEP.	Python 3.12, pinned OCP 7.9.3.1.1, and 85 repository tests passed with Boolean, clearance, section, tessellation, and visual acceptance.	Real visual geometry remained engineering-review evidence, not automatic production approval.
18	Build the Edit-Regenerate-Revalidate Workflow	Complete #29	Implemented move, resize, suppress, replace, restore, and parameter edits as revisioned commands that regenerate geometry, rerun validation, revoke stale approval, preserve history, compare revisions, and fail closed for unsupported edits.	Recorded acceptance included 88 passed/4 skipped and later repaired merge evidence referenced 91 passing tests; revision and approval-revocation paths passed.	Only explicitly supported engineering edits are allowed; source CAD remains immutable.
19	Deepen Weld-Fixture and Automation Workflow	Complete #39	Added editable weld, tack, clamp, release, loading, and unloading sequences; heat, distortion, spatter, and restricted zones; and shared torch, hand, operator, robot, cobot, load, and unload envelopes.	Pinned-OCF acceptance and workflow regressions passed with findings linked to workflow objects, collided geometry, and authoritative validation.	No thermal simulation, force simulation, weld-quality certification, or robot kinematic planning.
20	Harden Projects, Packaging, and Release Operations	Complete #40	Added project-v2 compatibility, atomic saves, autosave/recovery, diagnostics, isolated preferences, gated fabrication export, reproducible install/update, synthetic performance evidence, release manifests, sensitive-path rejection, and signing procedure.	Kernel acceptance, migration, autosave/recovery, preferences, large-assembly performance, manifest safety, sensitive-path rejection, and repository validation passed.	A controlled release procedure is not itself a production-approved release.
21	Generate Complete Fixture Structures	Complete #43	Added deterministic baseplate and welded-frame structures with connected members, load paths, sizing assumptions, evidence, and fail-closed findings integrated into concepts, regeneration, validation, persistence, and review.	9 focused structural tests and 110 full-suite tests passed; backlog, schema, compilation, secret scan, and real OCP proof passed.	No structural certification, final fabrication geometry, thermal simulation, robot planning, safety certification, or production approval.
22	Optimize Locator, Support, Stop, and Clamp Placement	Complete #44	Added datum-candidate ranking and editable placement contracts composing the six-DOF solver, access findings, tooling, weld intent, and structures while preserving alternatives, confidence, assumptions, digests, and evidence.	10 focused placement tests and 121 full-suite tests passed; compilation, backlog, schema, secret scan, and OCP proof passed.	No final B-Rep tooling, structural/thermal simulation, robot motion planning, safety certification, or production approval.
23	Produce Manufacturing-Ready Fixture Geometry	Complete #45	Added stable manufacturing-component contracts for real OCP/B-Rep fixture solids, deterministic per-component STEP and planar DXF exports, and source/component connectivity/holes/fit/mount/interface validation.	7 focused tests and 128 full-suite tests passed; compilation, backlog, schema, OCP proof, secret scan, STEP reimport, topology, byte determinism, and DXF checks passed.	Engineering-review exports only; no final drawings, complete tolerance stack, structural/safety certification, vendor approval, or production release.

M	Milestone	Status / PR	Scope Summary	Testing / Evidence	Boundary
24	Generate Fixture Drawings and Documentation	Complete #46	Generated typed assembly, exploded, component, purchased-reference, and evidence-backed detail sheets with dimensions, annotations, hole tables, BOM, revision blocks, findings, deterministic PDF, manifest, and drawing BOM.	8 focused tests and 137 full-suite tests passed; compileall, schema, backlog, secret scan, and OCP proof passed; a representative PDF had 22 pages and explicit review-only labels.	Qualified drawing review remained required; no production approval or certification.
25	Optimize Cost, Volume, and Manufacturability	Complete #47	Added deterministic cost, volume, manufacturability, alternative, recommendation, and evidence contracts with explicit material, process, engineering, programming, build, commissioning, maintenance, and replacement assumptions.	7 focused optimization tests and 147 full-suite tests passed; compile, schema, backlog, secret scan, and OCP proof passed; scenarios covered 1, 25, 250, and 1000 units.	Engineering estimate only; not supplier quotation, certified material takeoff, production budget, safety approval, or production release.

3. Detailed Milestone Records

Milestone 01 - Establish the Runnable Technical Baseline

Status	Complete	Controlling PR	#2
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Engineering scope and significance

Created the dependency-free geometry proof, explicit millimetre units, transforms, intersection and clearance checks, deterministic neutral serialization, CI entry points, and the first CAD-neutral architecture boundary.

Validation and source evidence

Baseline geometry and unit tests passed; the proof deliberately used bounded review geometry rather than claiming production B-Rep behavior.

Known limitations and protected boundaries

No STEP parser, real topology, production fixture geometry, certification, or production release.

Completion record

Milestone 01 is recorded as Complete because its controlling implementation was merged through PR #2. The original pull request, commit history, CI evidence, test records, and qualified engineering review remain authoritative.

Milestone 02 - Import STEP Assemblies into a Normalized Product Model

Status	Complete	Controlling PR	#3
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Engineering scope and significance

Added immutable CAD-neutral assembly hierarchy, repeated instances, nested transforms, units, bounds, topology summaries, source hashing, and explicit import failures.

Validation and source evidence

Hierarchy, transform, unit, repeated-instance, source-hash, and failure-path coverage passed.

Known limitations and protected boundaries

The normalized model did not modify source CAD or claim a full ISO 10303 implementation.

Completion record

Milestone 02 is recorded as Complete because its controlling implementation was merged through PR #3. The original pull request, commit history, CI evidence, test records, and qualified engineering review remain authoritative.

Milestone 03 - Build the Engineering-Annotation Workflow

Stops	Complete	Generating PR	#4
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Engineering scope and significance

Created source-bound annotations for build orientation, loading direction, process, quantity, critical characteristics, locating permissions, forbidden contacts, weld joints, shop constraints, and visible assumptions.

Validation and source evidence

Deterministic round-trip, source binding, reference validation, and explicit-unknown behavior passed.

Known limitations and protected boundaries

Annotations describe manufacturing intent; they do not prove or approve a fixture.

Completion record

Milestone 03 is recorded as Complete because its controlling implementation was merged through PR #4. The original pull request, commit history, CI evidence, test records, and qualified engineering review remain authoritative.

Milestone 04 - Generate Baseplate, Supports, Stops, and Locator Primitives

Stops	Complete	Generating PR	#5
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Engineering scope and significance

Generated traceable parameterized baseplates, supports, stops, pin envelopes, and locator primitives with initial overlap, forbidden-contact, missing-intent, and trapped-part findings.

Validation and source evidence

Primitive identity, parameter, overlap, forbidden-contact, missing-intent, and trapped-part coverage passed.

Known limitations and protected boundaries

Proof primitives only; no complete structure, structural adequacy, or production-ready geometry.

Completion record

Milestone 04 is recorded as Complete because its controlling implementation was merged through PR #5. The original pull request, commit history, CI evidence, test records, and qualified engineering review remain authoritative.

Milestone 05 - Create and Rank Complete Fixture Concepts

Stops	Complete	Controlling PR	#6
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Engineering scope and significance

Produced deterministic alternatives for minimum cost, fast loading, and high repeatability with locating/clamping rationale, warnings, score components, and copy-on-write corrections.

Validation and source evidence

Concept generation, ranking, score decomposition, rationale, alternatives, and correction behavior passed.

Known limitations and protected boundaries

Relative engineering ranking only; not supplier quotation, process certification, or production approval.

Completion record

Milestone 05 is recorded as Complete because its controlling implementation was merged through PR #6. The original pull request, commit history, CI evidence, test records, and qualified engineering review remain authoritative.

Milestone 06 - Model Weld, Operator, and Robot Access

Stops	Complete	Controlling PR	#7
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Engineering scope and significance

Added weld-approach, manual-tool, robot, operator, loading, and unloading envelopes with deterministic obstruction findings and visible uncertainty.

Validation and source evidence

Envelope generation, obstruction findings, incomplete-evidence handling, and traceability passed.

Known limitations and protected boundaries

No robot path planning, ergonomic certification, or weld-process approval.

Completion record

Milestone 06 is recorded as Complete because its controlling implementation was merged through PR #7. The original pull request, commit history, CI evidence, test records, and qualified engineering review remain authoritative.

Milestone 07 - Add Standard Clamp and Tooling Libraries

Complete	Open PR	#8
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Engineering scope and significance

Implemented vendor-neutral clamp, pin, and rest contracts and deterministic selection using force, stroke, mounting, envelope, licensing, and review limitations.

Validation and source evidence

Library identity, suitability, force/stroke, mounting, envelope, fallback, and licensing metadata passed.

Known limitations and protected boundaries

Review data only; not certified supplier selection or commercial-CAD equivalence.

Completion record

Milestone 07 is recorded as Complete because its controlling implementation was merged through PR #8. The original pull request, commit history, CI evidence, test records, and qualified engineering review remain authoritative.

Milestone 08 - Export a Fabrication-Ready Review Package

Complete	Open PR	#9
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Engineering scope and significance

Added deterministic review-package export with STEP-shaped proof geometry, DXF profiles, BOM, setup instructions, manifest, validation findings, units, revision, and engineering-review labels.

Validation and source evidence

Package determinism, manifest linkage, units, revision identity, and review-status coverage passed.

Known limitations and protected boundaries

Review package only; proof outputs were not released production drawings.

Completion record

Milestone 08 is recorded as Complete because its controlling implementation was merged through PR #9. The original pull request, commit history, CI evidence, test records, and qualified engineering review remain authoritative.

Milestone 09 - Capture Engineer Corrections and Reusable Knowledge

STATUS	Complete	Pending PR	#10
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Engineering scope and significance

Added attributable copy-on-write corrections, local private storage, sanitized training export, decision gating, and protections against silently promoting preferences into universal rules.

Validation and source evidence

Correction attribution, revision, privacy, sanitized export, and promotion-gating coverage passed.

Known limitations and protected boundaries

No autonomous engineering-rule rewriting.

Completion record

Milestone 09 is recorded as Complete because its controlling implementation was merged through PR #10. The original pull request, commit history, CI evidence, test records, and qualified engineering review remain authoritative.

Milestone 10 - Add CAD-Specific Connectors

STATUS	Complete	Pending PR	#11
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Engineering scope and significance

Established optional thin CAD connectors around the neutral model, including neutral STEP translation, review-package export, conservative SOLIDWORKS probing, and approval-gated destructive operations.

Validation and source evidence

Connector routing, capability probing, neutral handoff, failure behavior, and approval gates passed.

Known limitations and protected boundaries

The engineering core remained independent of SOLIDWORKS and every single CAD vendor.

Completion record

Milestone 10 is recorded as Complete because its controlling implementation was merged through PR #11. The original pull request, commit history, CI evidence, test records, and qualified engineering review remain authoritative.

Milestone 11 - Integrate a Real Geometry Kernel

Status	Complete	Controlling PR	#12
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Engineering scope and significance

Integrated pinned cadquery-ocp behind the CAD-neutral kernel boundary with real STEP import/export, XCAF hierarchy, topology, stable references, Booleans, distance, clearance, malformed-input handling, source immutability, and licensing records.

Validation and source evidence

Real-kernel import, topology, references, Booleans, distance, clearance, malformed input, source immutability, and licensing passed; superseded PR #13 was not controlling evidence.

Known limitations and protected boundaries

Kernel capability does not certify fixture geometry or manufacturing readiness.

Completion record

Milestone 11 is recorded as Complete because its controlling implementation was merged through PR #12. The original pull request, commit history, CI evidence, test records, and qualified engineering review remain authoritative.

Milestone 12 - Complete the Deterministic Locating and Constraint Solver

Status	Complete	Controlling PR	#14
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Engineering scope and significance

Implemented six-degree-of-freedom locating analysis using contact points, normals, roles, rigid-body constraint rows, rank analysis, and physical DOF classification.

Validation and source evidence

Rank, DOF classification, invalid-reference, redundancy, underconstraint, and recommendation-blocking coverage passed.

Known limitations and protected boundaries

Rigid-body locating analysis only; no elastic deformation, complete tolerance stack, or structural simulation.

Completion record

Milestone 12 is recorded as Complete because its controlling implementation was merged through PR #14. The original pull request, commit history, CI evidence, test records, and qualified engineering review remain authoritative.

Milestone 13 - Complete Weld-Fixture Engineering Rules

STATUS	Complete	Controlling PR	#15
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Engineering scope and significance

Added deterministic weld-process reasoning for process, direction, sequence, heat input, distortion, tack, release, support placement, and clamp-force direction with traceable rule identities.

Validation and source evidence

Clamp direction, tack/release, sequence, evidence, and rule-traceability coverage passed.

Known limitations and protected boundaries

No thermal FEA, weld-quality certification, or WPS approval.

Completion record

Milestone 13 is recorded as Complete because its controlling implementation was merged through PR #15. The original pull request, commit history, CI evidence, test records, and qualified engineering review remain authoritative.

Milestone 14 - Establish the Manufacturing-Aware Geometry Foundation

STATUS	Complete	Controlling PR	#16/#17/#19/#22
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Engineering scope and significance

Added manufacturing method, material, thickness, fit, clearance, allowance, interface, and operation metadata; deterministic cut-operation planning; source binding; feature-order validation; and STEP/DXF parity.

Validation and source evidence

STEP/DXF parity, wrong source hash, reordered features, malformed outputs, substitution, and pinned-OCP acceptance infrastructure passed.

Known limitations and protected boundaries

Foundation closure did not itself claim application-grade visual acceptance; that was completed under Milestone 17.

Completion record

Milestone 14 is recorded as Complete because its controlling implementation was merged through PR #16/#17/#19/#22. The original pull request, commit history, CI evidence, test records, and qualified engineering review remain authoritative.

Milestone 15 - Build the Full Deterministic Validation Pipeline

Status	Complete	Controlling PR	#21
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Engineering scope and significance

Unified geometry, locating, access, tooling, tolerance, and manufacturing evidence into a versioned ValidationResult that gates export and fails closed for missing, invalid, mismatched, unitless, unsupported, or empty evidence.

Validation and source evidence

Identity matching, evidence digest, intended interfaces, unrelated collisions, locating/clamp adequacy, provisional findings, and export blocking passed.

Known limitations and protected boundaries

Validation supports qualified review; it is not certification.

Completion record

Milestone 15 is recorded as Complete because its controlling implementation was merged through PR #21. The original pull request, commit history, CI evidence, test records, and qualified engineering review remain authoritative.

Milestone 16 - Build the First Serious Visual Engineering Application

Status	Complete	Controlling PR	#24
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Engineering scope and significance

Added a local engineering application for STEP import, rotatable review, layers, assumptions, findings, corrections, approval decisions, and neutral project save/reload with stale and invalid approval blocking.

Validation and source evidence

Annotation preservation, immutable source identity, evidence persistence, tamper detection, unsafe approval blocking, edit revocation, and unknown-feature rejection passed.

Known limitations and protected boundaries

Initial display used normalized review geometry; production-quality B-Rep display followed in Milestone 17.

Completion record

Milestone 16 is recorded as Complete because its controlling implementation was merged through PR #24. The original pull request, commit history, CI evidence, test records, and qualified engineering review remain authoritative.

Milestone 17 - Prove and Expose Real-Kernel Geometry

Complete	Open PR	#28
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Engineering scope and significance

Exposed real product and fixture B-Rep geometry with tessellation, selectable faces/edges, sections, feature findings, collision highlighting, wireframe, transparency, layers, regeneration, and reconstruction from embedded STEP.

Validation and source evidence

Python 3.12, pinned OCP 7.9.3.1.1, and 85 repository tests passed with Boolean, clearance, section, tessellation, and visual acceptance.

Known limitations and protected boundaries

Real visual geometry remained engineering-review evidence, not automatic production approval.

Completion record

Milestone 17 is recorded as Complete because its controlling implementation was merged through PR #28. The original pull request, commit history, CI evidence, test records, and qualified engineering review remain authoritative.

Milestone 18 - Build the Edit-Regenerate-Revalidate Workflow

Complete	Open PR	#29
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Engineering scope and significance

Implemented move, resize, suppress, replace, restore, and parameter edits as revisioned commands that regenerate geometry, rerun validation, revoke stale approval, preserve history, compare revisions, and fail closed for unsupported edits.

Validation and source evidence

Recorded acceptance included 88 passed/4 skipped and later repaired merge evidence referenced 91 passing tests; revision and approval-revocation paths passed.

Known limitations and protected boundaries

Only explicitly supported engineering edits are allowed; source CAD remains immutable.

Completion record

Milestone 18 is recorded as Complete because its controlling implementation was merged through PR #29. The original pull request, commit history, CI evidence, test records, and qualified engineering review remain authoritative.

Milestone 19 - Deepen Weld-Fixture and Automation Workflow

Status	Complete	Controlling PR	#39
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Engineering scope and significance

Added editable weld, tack, clamp, release, loading, and unloading sequences; heat, distortion, spatter, and restricted zones; and shared torch, hand, operator, robot, cobot, load, and unload envelopes.

Validation and source evidence

Pinned-OCP acceptance and workflow regressions passed with findings linked to workflow objects, collided geometry, and authoritative validation.

Known limitations and protected boundaries

No thermal simulation, force simulation, weld-quality certification, or robot kinematic planning.

Completion record

Milestone 19 is recorded as Complete because its controlling implementation was merged through PR #39. The original pull request, commit history, CI evidence, test records, and qualified engineering review remain authoritative.

Milestone 20 - Harden Projects, Packaging, and Release Operations

Status	Complete	Controlling PR	#40
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Engineering scope and significance

Added project-v2 compatibility, atomic saves, autosave/recovery, diagnostics, isolated preferences, gated fabrication export, reproducible install/update, synthetic performance evidence, release manifests, sensitive-path rejection, and signing procedure.

Validation and source evidence

Kernel acceptance, migration, autosave/recovery, preferences, large-assembly performance, manifest safety, sensitive-path rejection, and repository validation passed.

Known limitations and protected boundaries

A controlled release procedure is not itself a production-approved release.

Completion record

Milestone 20 is recorded as Complete because its controlling implementation was merged through PR #40. The original pull request, commit history, CI evidence, test records, and qualified engineering review remain authoritative.

Milestone 21 - Generate Complete Fixture Structures

Complete	Complete	Complete PR	#43
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Engineering scope and significance

Added deterministic baseplate and welded-frame structures with connected members, load paths, sizing assumptions, evidence, and fail-closed findings integrated into concepts, regeneration, validation, persistence, and review.

Validation and source evidence

9 focused structural tests and 110 full-suite tests passed; backlog, schema, compilation, secret scan, and real OCP proof passed.

Known limitations and protected boundaries

No structural certification, final fabrication geometry, thermal simulation, robot planning, safety certification, or production approval.

Completion record

Milestone 21 is recorded as Complete because its controlling implementation was merged through PR #43. The original pull request, commit history, CI evidence, test records, and qualified engineering review remain authoritative.

Milestone 22 - Optimize Locator, Support, Stop, and Clamp Placement

Complete	Complete	Complete PR	#44
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Engineering scope and significance

Added datum-candidate ranking and editable placement contracts composing the six-DOF solver, access findings, tooling, weld intent, and structures while preserving alternatives, confidence, assumptions, digests, and evidence.

Validation and source evidence

10 focused placement tests and 121 full-suite tests passed; compilation, backlog, schema, secret scan, and OCP proof passed.

Known limitations and protected boundaries

No final B-Rep tooling, structural/thermal simulation, robot motion planning, safety certification, or production approval.

Completion record

Milestone 22 is recorded as Complete because its controlling implementation was merged through PR #44. The original pull request, commit history, CI evidence, test records, and qualified engineering review remain authoritative.

Milestone 23 - Produce Manufacturing-Ready Fixture Geometry

Status	Complete	Controlling PR	#45
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Engineering scope and significance

Added stable manufacturing-component contracts for real OCP/B-Rep fixture solids, deterministic per-component STEP and planar DXF exports, and source/component/connectivity/hole/fit/mount/interface validation.

Validation and source evidence

7 focused tests and 128 full-suite tests passed; compileall, backlog, schema, OCP proof, secret scan, STEP reimport, topology, byte determinism, and DXF checks passed.

Known limitations and protected boundaries

Engineering-review exports only; no final drawings, complete tolerance stack, structural/safety certification, vendor approval, or production release.

Completion record

Milestone 23 is recorded as Complete because its controlling implementation was merged through PR #45. The original pull request, commit history, CI evidence, test records, and qualified engineering review remain authoritative.

Milestone 24 - Generate Fixture Drawings and Documentation

Status	Complete	Controlling PR	#46
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Engineering scope and significance

Generated typed assembly, exploded, component, purchased-reference, and evidence-backed detail sheets with dimensions, annotations, hole tables, BOM, revision blocks, findings, deterministic PDF, manifest, and drawing BOM.

Validation and source evidence

8 focused tests and 137 full-suite tests passed; compileall, schema, backlog, secret scan, and OCP proof passed; a representative PDF had 22 pages and explicit review-only labels.

Known limitations and protected boundaries

Qualified drawing review remained required; no production approval or certification.

Completion record

Milestone 24 is recorded as Complete because its controlling implementation was merged through PR #46. The original pull request, commit history, CI evidence, test records, and qualified engineering review remain authoritative.

Milestone 25 - Optimize Cost, Volume, and Manufacturability

Status	Complete	Controlling PR	#47
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Engineering scope and significance

Added deterministic cost, volume, manufacturability, alternative, recommendation, and evidence contracts with explicit material, process, engineering, programming, build, commissioning, maintenance, and replacement assumptions.

Validation and source evidence

7 focused optimization tests and 147 full-suite tests passed; compile, schema, backlog, secret scan, and OCP proof passed; scenarios covered 1, 25, 250, and 1000 units.

Known limitations and protected boundaries

Engineering estimate only; not supplier quotation, certified material takeoff, production budget, safety approval, or production release.

Completion record

Milestone 25 is recorded as Complete because its controlling implementation was merged through PR #47. The original pull request, commit history, CI evidence, test records, and qualified engineering review remain authoritative.

4. Evidence and Validation Record

Controlling evidence consists of merged pull requests, commit history, automated and hosted validation, real-kernel proof, Windows visual evidence where applicable, architecture decisions, project records, roadmap records, and explicit human-review boundaries. Exact missing historical values are not invented.

5. Documentation Correction Note

This audited edition replaces the earlier under-filled print layout. It restores dense milestone tracking, engineering scope, validation evidence, limitations, closeout, and content-driven pagination while preserving repository history as the source of truth.

6. Volume Closeout

Volume 1 closes at Milestone 25 because the first complete deterministic engineering foundation was established: imported assemblies could be interpreted, annotated, fixtured, validated, represented as manufacturing components, documented, and compared economically while preserving source-CAD immutability and human approval.

7. Controlled Handoff

Volume 2 begins at Milestone 26 and records the transition from the deterministic foundation into the local engineering workbench, unified application, interactive workflow, branding, real fixture-build workflows, and guided AI assistance.

8. Future Documentation Rule

- Planning -> Implementation -> Testing -> Documentation.
- Never mark an open or unmerged milestone complete.
- Update controlled binder batches while evidence is fresh.
- Record PR, validation evidence, limitations, and the next handoff.
- Pass results, not noise; never reconstruct missing facts by guessing.



FXD Engineering Project Binder

Volume 2 - Milestones 26-31

Audited engineering record based on the approved documentation-recovery structure

Project	FXD - Intelligent Industrial Fixture Design
Repository	kool1160/fxd-fixture-design
Document Class	Engineering development history / professional project binder
Volume Status	Closed audited record through Milestone 31
Audit Basis	Merged pull requests, tests, CI, project law, architecture, decision, and project records
Runtime Changes	None - documentation only

AI proposes. Engineering validates.

The software should think like an experienced fixture engineer.

1. Executive Summary

Volume 2 records the transformation of FXD from a completed deterministic fixture-engineering foundation into a serious local engineering application with real STEP viewing, a unified PySide6/VTK workbench, interactive engineering workflows, controlled branding, real fixture-construction geometry, and the advisory AI Fixture Engineer with guided validation.

Governing principles

- Source customer CAD remains immutable.
- Deterministic geometry, mathematics, and engineering rules remain authoritative.
- AI is advisory and may not approve a fixture.
- Every recommendation and validation result must remain explainable and traceable.
- Manufacturing practicality and qualified human approval remain mandatory.

2. Master Milestone Recovery Tracker

Exact historical test counts are reported only when the controlling repository evidence states them.

M	Milestone	Status / PR	Scope Summary	Testing / Evidence	Boundary
26	Complete the Local Engineering Workbench	Complete #48	Repaired OCCT one-based triangle indexing and added persistent VTK actors, shaded defaults, navigation, diagnostics, XCAF color preservation, and honest real-geometry fallback.	5 focused workbench tests and 154 full-suite tests passed; kernel proof, backlog, compileall, diff checks, and controlled STEP evidence passed.	Render benchmarks were not visible-GUI FPS claims; engineering review only.
27	Unify the FXD Engineering Workbench	Complete #49	Replaced the launched Tk shell with a PySide6 workbench and supervised native VTK worker, adding engineering tree, properties/findings, layers, review decisions, renderer controls, diagnostics, benchmarks, selection, and visibility.	15 focused tests and 164 full-suite tests passed; CI contract, full CI, OCP proof, compileall, backlog, schema, diff checks, and Windows acceptance passed.	Windows child-window embedding is not cross-platform acceptance; provisional evidence cannot be labeled real source geometry.
28	Expose the Interactive Fixture Engineering Workflow	Complete #50	Exposed process intent, exact OCP face annotations, private tooling metadata, deterministic background analysis, concept comparison, supported corrections, revision restoration, findings review, persistence, and gated evidence export.	34 focused workflow/workbench tests and 190 full-suite tests passed; compileall, schema, backlog, CI, secret scan, OCP proof, and Windows acceptance passed.	Concept evidence remained visibly provisional; no paid runtime AI, supplier scraping, thermal simulation, robot planning, safety certification, or production approval.
29	Implement the Desktop UI and Branding System	Complete #51	Integrated UI and Branding Kit v1.1 with design tokens, palette, QSS, approved assets, semantic widgets, source identity, workflow rail, renderer health, validation/approval gates, responsive docks, and layout persistence.	30 focused branding/workbench tests and 201 full-suite tests passed; compileall, schema, backlog, launcher, secret scan, OCP proof, full CI, and branding-manifest verification passed.	Branding does not alter engineering rules, source identity, validation, or approval authority.
30	Generate Real Fixture Geometry and Tack/Location Workflows	Complete #52	Added deterministic real OCP fixture-construction evidence for full-weld, tack/location, assembly, inspection, checking, rework, automation, and combined fixture purposes with lifecycle, Cleco, holes, tabs, poka-yoke, BOM, nest, sequence, authority, validation, comparison, authoring, and schema-v4 persistence.	16 focused tests and 224 full-suite tests passed; compileall, backlog, schema, OCP proof, full CI, and Windows real-geometry evidence passed.	No structural adequacy, clamp-force calculation, thermal simulation, robot planning, automated nesting, safety certification, weld-procedure approval, or production release.
31	AI Fixture Engineer and Guided Validation	Complete #53	Added the provider-neutral fxd-fixture-proposal-v1 contract, environment-only adapter, deterministic baseline, output quarantine, proposal workflow, intent confirmation, audited recommendations, evidence highlights, guided validation/correction routing, provenance, staleness, tooling evidence, and schema-v5 persistence.	17 AI-core, 45 Qt/workbench, 14 persistence/export, 36 orientation/interactive/kernel, and 278 full-suite tests passed; compileall, diff, launcher, CI, hosted OCP acceptance, OCP proof, and Windows walkthrough passed.	AI never approves fixtures; deterministic validation and human approval remain authoritative and fail closed.

3. Detailed Milestone Records

Milestone 26 - Complete the Local Engineering Workbench

Status	Complete	Controlling PR	#48
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Engineering scope and significance

Repaired OCCT one-based triangle indexing and added persistent VTK actors, shaded defaults, navigation, diagnostics, XCAF color preservation, and honest real-geometry fallback.

Validation and source evidence

5 focused workbench tests and 154 full-suite tests passed; kernel proof, backlog, compileall, diff checks, and controlled STEP evidence passed.

Known limitations and protected boundaries

Render benchmarks were not visible-GUI FPS claims; engineering review only.

Completion record

Milestone 26 is recorded as Complete because its controlling implementation was merged through PR #48. The original pull request, commit history, CI evidence, test records, and qualified engineering review remain authoritative.

Milestone 27 - Unify the FXD Engineering Workbench

Status	Complete	Controlling PR	#49
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Engineering scope and significance

Replaced the launched Tk shell with a PySide6 workbench and supervised native VTK worker, adding engineering tree, properties/findings, layers, review decisions, renderer controls, diagnostics, benchmarks, selection, and visibility.

Validation and source evidence

15 focused tests and 164 full-suite tests passed; CI contract, full CI, OCP proof, compileall, backlog, schema, diff checks, and Windows acceptance passed.

Known limitations and protected boundaries

Windows child-window embedding is not cross-platform acceptance; provisional evidence cannot be labeled real source geometry.

Completion record

Milestone 27 is recorded as Complete because its controlling implementation was merged through PR #49. The original pull request, commit history, CI evidence, test records, and qualified engineering review remain authoritative.

Milestone 28 - Expose the Interactive Fixture Engineering Workflow

Status	Complete	Controlling PR	#50
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Engineering scope and significance

Exposed process intent, exact OCP face annotations, private tooling metadata, deterministic background analysis, concept comparison, supported corrections, revision restoration, findings review, persistence, and gated evidence export.

Validation and source evidence

34 focused workflow/workbench tests and 190 full-suite tests passed; compileall, schema, backlog, CI, secret scan, OCP proof, and Windows acceptance passed.

Known limitations and protected boundaries

Concept evidence remained visibly provisional; no paid runtime AI, supplier scraping, thermal simulation, robot planning, safety certification, or production approval.

Completion record

Milestone 28 is recorded as Complete because its controlling implementation was merged through PR #50. The original pull request, commit history, CI evidence, test records, and qualified engineering review remain authoritative.

Milestone 29 - Implement the Desktop UI and Branding System

Status	Complete	Controlling PR	#51
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Engineering scope and significance

Integrated UI and Branding Kit v1.1 with design tokens, palette, QSS, approved assets, semantic widgets, source identity, workflow rail, renderer health, validation/approval gates, responsive docks, and layout persistence.

Validation and source evidence

30 focused branding/workbench tests and 201 full-suite tests passed; compileall, schema, backlog, launcher, secret scan, OCP proof, full CI, and branding-manifest verification passed.

Known limitations and protected boundaries

Branding does not alter engineering rules, source identity, validation, or approval authority.

Completion record

Milestone 29 is recorded as Complete because its controlling implementation was merged through PR #51. The original pull request, commit history, CI evidence, test records, and qualified engineering review remain authoritative.

Milestone 30 - Generate Real Fixture Geometry and Tack/Location Workflows

Status	Complete	Controlling PR	#52
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Engineering scope and significance

Added deterministic real OCP fixture-construction evidence for full-weld, tack/location, assembly, inspection, checking, rework, automation, and combined fixture purposes with lifecycle, Cleco, holes, tabs, poka-yoke, BOM, nest, sequence, authority, validation, comparison, authoring, and schema-v4 persistence.

Validation and source evidence

16 focused tests and 224 full-suite tests passed; compileall, backlog, schema, OCP proof, full CI, and Windows real-geometry evidence passed.

Known limitations and protected boundaries

No structural adequacy, clamp-force calculation, thermal simulation, robot planning, automated nesting, safety certification, weld-procedure approval, or production release.

Completion record

Milestone 30 is recorded as Complete because its controlling implementation was merged through PR #52. The original pull request, commit history, CI evidence, test records, and qualified engineering review remain authoritative.

Milestone 31 - AI Fixture Engineer and Guided Validation

Status	Complete	Controlling PR	#53
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Engineering scope and significance

Added the provider-neutral fxd-fixture-proposal-v1 contract, environment-only adapter, deterministic baseline, output quarantine, proposal workflow, intent confirmation, audited recommendations, evidence highlights, guided validation/correction routing, provenance, staleness, tooling evidence, and schema-v5 persistence.

Validation and source evidence

17 AI-core, 45 Qt/workbench, 14 persistence/export, 36 orientation/interactive/kernel, and 278 full-suite tests passed; compileall, diff, launcher, CI, hosted OCP acceptance, OCP proof, and Windows walkthrough passed.

Known limitations and protected boundaries

AI never approves fixtures; deterministic validation and human approval remain authoritative and fail closed.

Completion record

Milestone 31 is recorded as Complete because its controlling implementation was merged through PR #53. The original pull request, commit history, CI evidence, test records, and qualified engineering review remain authoritative.

4. Evidence and Validation Record

Controlling evidence consists of merged pull requests, commit history, automated and hosted validation, real-kernel proof, Windows visual evidence where applicable, architecture decisions, project records, roadmap records, and explicit human-review boundaries. Exact missing historical values are not invented.

5. Documentation Correction Note

This audited edition replaces the earlier under-filled print layout. It restores dense milestone tracking, engineering scope, validation evidence, limitations, closeout, and content-driven pagination while preserving repository history as the source of truth.

6. Volume Closeout

Volume 2 closes at Milestone 31. FXD now has a professional Windows engineering workbench, real source-geometry review, interactive intent and correction workflows, deterministic real fixture-build geometry, governed persistence, and an advisory AI proposal layer subordinate to validation and human authority.

7. Controlled Handoff

Milestone 32 begins the next fixture-family expansion phase, starting with governed multi-station weld-fixture synthesis. It belongs to the next binder volume or addendum.

8. Future Documentation Rule

- Planning -> Implementation -> Testing -> Documentation.
- Never mark an open or unmerged milestone complete.
- Update controlled binder batches while evidence is fresh.
- Record PR, validation evidence, limitations, and the next handoff.
- Pass results, not noise; never reconstruct missing facts by guessing.